WUHAN UNIVERSITY

WEAK EMERGENCE-BASED NON-REDUCTIVE PHYSICALISM

A Thesis submitted in partial satisfaction of the requirements
for the degree Bachelor of Arts

in

Philosophy

by

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Defended on

May 9th, 2023

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DEDICATION

For David Kellogg Lewis
The one, the only and the forever

ABSTRACT

The notion of (weak) metaphysical emergence proposed by Jessica M. Wilson hopefully enriches our inter-level metaphysical toolbox. However, there are criticisms about Wilson's allegedly overstated theoretical neutrality, when applying her theory to the formulation of non-reductive physicalism. I start this project by arguing that Wilsonian weak emergentists cannot be neutral to the nature of properties. Later, I propose the dual-aspect model of properties—that a property is represented by both its quiddity and its causal powers—to help Wilsonian emergentists i) differentiate weak emergence from strong emergence qualitatively, ii) defend the non-reductiveness of emergent-level properties, and iii) explain why many emergent-level laws are *ceteris paribus* laws. To summarize, I believe the dual-aspect model of properties can provide Wilsonian emergentism an ontological foundation.

Keywords: Wilsonian emergentism; weak emergence; non-reductive physicalism; dispositions; quiddities; the dual-aspect model of properties; *ceteris paribus* laws

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1 Metaphysical Emergence: Introductory Remarks

Reality seems to be multi-layered. There are lower-level entities like electrons, neutrons and quarks, and there are also higher-level entities like cells, organisms and living persons. What's more, a configuration of atoms exhibiting Brownian movement seems to bear a relation to the spatial-temporally co-located cell in some way. The sub-discipline of philosophy studying relations as such is known as inter-level metaphysics. Many candidates compete for a better characterization of inter-level metaphysical relations: supervenience, realization, philosophical grounding, and so on. About two decades ago, a competitor called emergence (re-)emerged and joined the game. The notion of emergence has a complicated history, affiliated with Darwinian evolutionism, scientific unificationalism, and many other intellectual movements (McLaughlin 2019, pp. 23–33; Macdonald and Macdonald 2019, pp. 195–203). But my focus in this paper is a rather recent conception of emergence—more precisely, weak emergence—which, as the predominant theory of emergence currently at issue, is first introduced in Wilson 2015 and further developed in Wilson 2021.

Emergence, as a metaphysical tool, can assumably be employed to characterize various relations: the relation between the mental and the physical, the relation between the moral and the natural, etc. Following Wilson, my focus in this paper is the formulation of non-reductive physicalism. The paper is so-focused for two reasons. First, the notion of weak emergence could offer a new approach to formulating non-reductive

physicalism which, if true, reconciles the physicalist manifesto (i.e., every existence is ultimately physical) and our mundane intuitions (e.g., there is something special about mentality)—or as Wilson (2021, p. 2) puts it, captures both *dependence* and *autonomy*. Second, the case study of formulating non-reductive physicalism can test the viability of emergence as a metaphysical tool itself. Its cons and pros could contribute to the evaluation and development of the industry in the broad picture. Later in this paper, whenever I consider a certain proposal, I start from a (non-reductive) physicalist perspective, unless otherwise notified.

Some preliminary remarks before I proceed. Wilsonian schema of metaphysical emergence is implicitly committed to three background presuppositions. The first presupposition is the layer-model of reality (Kim 1993, p. 337; Bennett 2017, p. 67). The general idea is that the reality is hierarchical, not flat, and is divided into distinct layers/levels (in ascending order of levels): the layer of particles, the layer of cells, and the layer of living persons, etc. Following Wilson (2021, pp. 30–32), I stipulate that lower-level entities are more fundamental than higher-level entities, and fundamental entities are "analogous to axioms in a theory—they are basic, they are 'all God had to do, or create'" (Wilson 2014, p. 560). The stipulation is made so that the No Fundamental Mentality constrain—i.e., the core physicalist constrain that Wilson and I advocate—can be accommodated[1]. Central to the No Fundamental Mentality constrain is the idea

[1] Pace Wilson (2021, pp. 22–24), I will not propose a definition of physicalism of any kind. On the contrary, the No Fundamental Mentality constrain is granted as a necessary condition of all versions of physicalism, though arguably not the only necessary condition of physicalism.

that

> "[C]ompositionally basic physical entities and features […] do not individually either possess or bestow mentality. … Note that this constraint, properly understood, attaches to the basic or comparatively noncomplex entities and features which are the targeted subject matter of physics, and so does not rule out configurations of physical entities or their features […] from individually having or bestowing mentality" (Wilson 2021, p. 23).

For convenience, I also grant that if something is fundamental, then it is ontologically primitive and not grounded in any other existence (where 'ground' should be understood as a respectful word in ordinary English, rather than a rigorously defined metaphysical notion).

The second presupposition is the Property Exemplification Account (henceforth PEA) of events (Kim 1976, p. 161), which basically says something like this. An event could be represented by a three-place ordered set, where the first element is an object/person, the second element is a functional property and the third element is time. Provided time is a parameter, and that an object/person at a certain time (or during a certain interval of time) can be considered as the instantiation of a host of properties, in the end, properties become the center of attention, despite the fact that properties, objects and events are apparently distinct categories of existence[2]. With the benefit of

[2] Notice that I do not intend to invite sub-debates of PEA like whether (mental) properties are constitutive properties of (mental) events, or whether a monadic event has just one constitutive property (Macdonald and Macdonald 2019, p. 197). All I need is the basic idea of PEA.

presupposing PEA, emergence-talks in this paper are property-concentrated—more accurately, objective property-concentrated[3].

The third presupposition is the Token Identity of Causal Powers. I start from token identity theory, a theoretical schema first introduced by reductive physicalists. Many non-reductive physicalists (like Jessica Wilson and arguably Sydney Shoemaker) also accept this schema, but they propose that the token identical relation obtains in causal powers of properties, not properties themselves[4]. A formal non-reductive physicalist definition of Token Identity of Causal Powers condition goes like this. Every token power of token mental property is identical with a token power of the token physical property from which the mental property emerges (Wilson 2021, p. 58).

I now motivate my project. My general position is pro-Wilsonian (weak) emergentism. But I agree that criticisms from Ney (2022) or Baysan (2022) targeting on Wilson's overstated theoretical neutrality pose substantial threats. Hence, the goal of this paper is in some sense to debug Wilson's theory. Here is a more detailed writing plan. In section 2, I first introduce Wilson's 2021 account of metaphysical emergence, and demonstrate three discontents of Wilson's account. Later in section 3, I propose the dual-aspect model of properties—that a property is represented by both its quiddity and its causal powers—which initially offers a qualitative distinction between weak and

[3] Or equivalently, feature-concentrated. In literature, authors like Jessica Wilson prefer the term 'feature' while authors like Jaegwon Kim (or me) prefer 'property'. And in what follows, I solely employ 'property', though I acknowledge that 'feature' and 'property' can be used interchangeably. Besides, here I also put the debate over qualia/psychological properties like redness, chilliness, etc. aside.

[4] Though dispositional essentialism about the nature of properties suggest that properties can be considered as profiles of causal powers. See my discussion in section 2.

strong emergence, and defends the non-reductiveness of emergent-level properties. I then in section 4 illustrate how the dual-aspect model of properties explains why many emergent-level laws are *ceteris paribus* laws. And I finally draw several periodic conclusions in section 5.

2 Wilson's Account of Emergence and its Discontents

There are two tasks for section 2—to introduce Wilson's account of weak emergence, and to illustrate three discontents of Wilson's account: the weakness in defending non-reductiveness, the absence of a quantitive distinction between weak and strong emergence, and the silence on the explanation of a difference in logical forms between emergent-level laws and base-level laws.

Wilson's account of emergence is defined in two other notions—co-temporal material dependence and causal powers. Here is a brief elaboration of each notion. Consider a macro-entity (like a tree) and some micro-entities (like some particles). Scientifically, a macro-entity is ultimately dependent on complex configurations of some micro-entities such that—at any given time, the macro-entity inherits its matter from some configuration of micro-entities; and over a given temporal interval, the macro-entity's some features (its states, properties, etc.) are at least partly a function of features of the underlying micro-entities' configuration (Wilson 2021, p. 1). Above is Wilson's conception of co-temporal material dependence[5]. And the notion of causal powers says

[5] Wilson's conception of co-temporal material dependence resembles but differs from the notion of metaphysical dependence. Following Wilson (2021, p. 150), I don't think an involvement in the nature

something like this. Causal powers are basically an entity's causal contributions made (or can be made) to bringing about an effect, in certain circumstances (Wilson 2021, p. 32; originally in Wilson 2015, p. 354). In other words, usually, an entity's causal powers are that entity's capacities to causally bring about certain states—usually certain states of some other entities[6]. More specifically, since this project is property-concentrated, in what follows, a property's causal powers are considered as its capacities to causally bring about (the instantiation of) other properties. For instance, suppose a (categorical) property, S, has the capacity to causally bring about (the instantiation of) another (categorical) property, M, then I grant that S has the causal power denoted by $D_{(S, M)}$[7].

I now put Wilson's account of weak emergence on the table, by contrast to Wilson's account of strong emergence:

> *Weak Emergence*. "What it is for token feature S to be Weakly metaphysically emergent from token feature P on a given occasion is for it to be the case, on that occasion, (i) that S cotemporally materially depends on P, and (ii) that S has a non-empty proper subset of the token powers had by P." (Wilson 2021, p. 72)

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of metaphysical dependence is necessary here.

[6] Here I do not distinguish causal powers from dispositions. And I do not presuppose any particular theory about the nature of causation, my focus is instead the nature of properties. But it does not mean I agree that Wilson(-ian emergentists) can be neutral to the nature of causation. See Baysan 2022, pp. 163–165 for the relevant argument.

[7] This notation is proposed for later discussion about laws in sec. 4. Though it is famously introduced by Alexander Bird, I do not presuppose any Bird-style conception here. In fact, here I do not presuppose any particular theory of the nature of dispositions, e.g., whether dispositions can be analyzed in simple conditionals, are dispositions primitive, or are dispositions intrinsic/extrinsic in nature, etc.

> *Strong Emergence*. "What it is for token feature S to be Strongly metaphysically emergent from token feature P on a given occasion is for it to be the case, on that occasion, (i) that S cotemporally materially depends on P, and (ii) that S has at least one token power not identical with any token power of P." (Wilson 2021, p. 53)

Wilson's accounts are about necessary conditions of weak/strong emergence. In addition, I want to emphasize another crucial point to the notion of emergence: its bruteness. Emergence itself is brute. We know emergence implies co-material dependence and set-membership of causal powers, but we do not know how and why emergence occurs. We know an (weakly) emergent property inherits some causal powers from its emergent base, but with respect to why inheriting these causal powers rather than some others, we are ignorant. It is understandable that some people find such bruteness dissatisfying, or even ad hoc. But to view it another way, the metaphysics of emergence is also a metaphysics with humility—since arguably, none of us is a perfect knower and human understanding on the whole is limited[8].

To proceed, I highlight two desiderata for the conception of weak emergence illustrated by Wilson as below:

> *Ontological non-reductiveness*. "[S]pecial-science entities also seem […] distinct from, and distinctively efficacious with respect to, the micro-configurations upon which they depend." (Wilson 2021, p. 1)

[8] Or in a slogan: Emergence is Mercy. This slogan originally comes from an Edward Spencer quotation from the British emergentist Samuel Alexander (1920, vol. 2, p. 391), which says, "highest God, that loves his creatures so, and all his works with mercy doth embrace."

> *Nomological autonomy*. "Special-science entities are taken to be governed by special-science laws (or regularities) […] that, on the face of it, are different from those governing even complex configurations of fundamental physical entities." (Wilson 2021, p. 4)

Besides, Wilson (2021, p. 18) also claims that in her theory "generally, no controversial theses pertaining to the nature of powers, properties, causation, or laws are presupposed"—i.e., Wilson's theory is proudly neutral to different conceptions of the nature of powers, properties, causation and laws. In what follows, I illustrate three discontents of Wilson's account. The first discontent is the weakness in defending the non-reductiveness of emergent-level properties[9].

My argument focuses on Proper Subset of Powers Condition, which says an emergent property's causal powers are a non-empty proper subset of the emergent base's causal powers. My core idea is that Proper Subset of Powers Condition *per se* cannot satisfy the ontological non-reductiveness desideratum. My line of reasoning goes like this. From Proper Subset of Powers Condition, it follows that token feature S's causal profile is non-identical with token feature P's causal profile, and more specifically, S's causal profile is included in P's causal profile. But non-identity is also compatible with reduction. Consider Sydney Shoemaker's theory of mereological realization (2007). As Kim (2010, pp. 110–112) points out, provided Proper Subset of Powers Condition (as

[9] Similar concerns can be found in Ney (2022, para. 5) or Baysan (2022, p. 159). I should clarify my objection is not raised as a repeat of other people's criticism, but raised as a preparation of my positive proposal, i.e., the dual-aspect model of properties.

is also given in Shoemaker 2007, ch. 2), and provided dispositional essentialism about the nature of properties—the position that all the significance there is to a property is its causal powers and a property is identified and individuated by its causal profile (as Shoemaker 1980 defends), the general picture is reductive in the sense that mental properties are ontologically just a part (i.e., a proper subset) of physical properties[10]. Hence, unlike Wilson's statement that "nothing in what follows requires accepting that features are exhaustively individuated by powers. Perhaps they are, perhaps they aren't; perhaps features are also or ultimately individuated by quiddities or other noncausal aspects of features" (2021, p. 33), to maintain a non-reductive spirit, a Wilsonian weak emergentist should reject dispositional essentialism about the nature of properties.

The weakness in defending non-reductiveness is the first discontent of Wilson's account of emergence. And there are two more discontents. The second discontent of Wilson's account is the absence of a quantitive distinction between weak and strong emergence. The original distinction in Wilson 2021 (p. 74) is mostly quantitive—whether a property is weakly or strongly emergent depends on two numbers: the number of its causal powers and the number of its emergent base's causal powers. I agree that this distinction captures some difference between weak and strong emergence, though in an arguably superficial way. Intuitively weak emergence should differ from

[10] Though Shoemaker's 2007 theory is arguably in favor of sematic irreducibility of mental terms—e.g., suppose the mental property *pain* is reducible to a physical property, *pain* still differs from that physical property in the sense that, in ordinary English, we do not have a term that denotes the physical property which is also a synonym for 'pain'—such kind of irreducibility can hardly satisfy the ontological non-reductiveness desideratum.

strong emergence in some more substantial and significant ways. But Wilson's account fails to characterize any way as such.

The third discontent of Wilson's account is its silence on the explanation of a difference in logical forms between emergent-level laws and base-level laws. To satisfy the nomological autonomy desideratum, Wilsonian emergentists need to answer two questions: in what way are emergent-level laws and base-level laws different, and why? Wilson points out one such way explicitly: emergent-level laws and base-level laws seem to have different explanatory domains. Generally, laws from different levels explain different phenomena, emergent-level laws cannot explain base-level phenomena, and *vice versa* (Wilson 2021, pp. 168–169). For instance, the explanation of the biological phenomenon of genetic inheritance should appeal to emergent-level laws like Mendel's Laws of Independent Assortment, rather than some base-level laws like Schrödinger's equation[11].

I buy into the difference in explanatory domain. Nevertheless, there is another philosophically interesting difference that is inadequately explored in Wilson 2021. This difference is: emergent-level laws and base-level laws seem to have different logical forms. Fundamental physical laws (like Laws of Special Relativity) are usually strict laws to which there is no exception. On the contrary, special-scientific laws (like the Mendel's Law of Segregation) are usually *ceteris paribus* laws to which an

[11] More generally, for any law to operate, some information (or formally put, some independent parameters) is needed to specify states relevant to an entity's properties. This line of reasoning is also known as Wilson's theory of degree of freedom, see Wilson 2021, pp. 30, 90 for details.

"interfering factors aside" clause have to be put (Wilson 2021, p. 35)[12]. Though Wilson notices the difference, she doesn't pursue the point. But Wilson's readers may reasonably be dissatisfied with her termination of inquiry.

3 The Dual-aspect Model of Properties

In section 3, I will advance a proposal that modifies Wilson's account of emergence, motivated by its discontents. My modification targets on one foundation of Wilson's account of emergence: her conception of the nature of properties. I have established in section 2 that, to accommodate non-reductive physicalism, Wilsonian weak emergentists need to presuppose a certain theory of the nature of properties. And this theory cannot be dispositional essentialism, which inclines Wilsonian weak emergentists to competing theories like categorical essentialism or hybrid views[13]. In what follows, I briefly review the debate over the nature of properties, and put forward a proposal called the dual-aspect model of properties.

One remark at first. My argumentative strategy differs from the one Wilson's objection-raisers (e.g., Umut Baysan) pursue. My goal is not to argue how Wilsonian weak

[12] Here I do not intend to invite debates over different readings of *ceteris paribus*-clause. I believe the "interfering factors aside" reading Wilson endorses is compatible with various readings like "other things being equal", "other things being normal", "other things being right", etc.

[13] Though dispositional essentialism, categorical essentialism, and hybrid views are most popular theories of the nature of properties, I do not want to exclude positions like there is no essence to properties at all (i.e., an anti-essentialist stance about the nature of properties, consider the bare particular anti-essentialism in Stalnaker 1979, p. 344). However, I doubt if there is any reason for Wilsonian emergentists to embrace non-popular views, especially when accepting a certain popular view proves (quite) beneficial to them.

emergentism would be problematic in the absence of certain theories. That is, I will not argue why Wilsonian weak emergentists *must* presuppose certain theories that are *best* for their interests. Instead, my strategy is gentler: I intend to show, if Wilsonian weak emergentists accept a certain theory, how their position can be better defended and how their theoretical desiderata can be better satisfied.

To start, I introduce the notion of quiddity, a notion constitutive to categorical essentialism about the nature of properties that Wilsonian emergentists may adopt. I will use 'quiddity', 'qualitative aspect' and 'categorical aspect' interchangeably henceforth. A quiddity is a property's categorical, non-relational and purely qualitative essence, i.e., the primitive/basic suchness[14]. Categorical essentialists about nature of properties believe that a property is solely identified and individuated by its quiddity. But dispositional essentialists disagree. Consider the quiddity *being an electron*. Dispositions like *having a spin quantum number of +½*, *having a rest mass of* 9.109×10^{-31} *kilograms* and *have an electric charge of 1–* usually come along with the quiddity *being an electron*. The disagreement between categorical- and dispositional essentialism about the nature of properties is, in some sense, about whether quiddities like *being an electron* are redundant since what scientifically signify are dispositions like *mass*, *charge* and

[14] The notion of quiddity is assumably a familiar concept for metaphysicians. Nevertheless, for readers who are unfamiliar with this notion, there is an illustrating example. *Being an electron* is a quiddity in the sense that: 1) it is, contra haecceity (i.e., the primitive/basic thisness), not a property of being identical with a particular individual (e.g., persons, physical objects, and events) but a property that many individuals (in this case, particles known as electrons) share, 2) it is not a property of being related to particular individuals (e.g., being an electron of a hydrogen atom), and 3) it is not a property of being identical with (or related in one way or another to) an extensionally defined set that has an individual among its members (e.g., if *being an American* is to be analyzed as a relation to a set of actual people and places, then it is not a quiddity) (Adams 1979, pp. 7–8).

spinning. Categorical essentialists say no, while dispositional essentialists say yes. But such a disagreement is not about compatibility: the idea that quiddities and causal powers can pair up is not esoteric for both parties. Inspired by this idea, many hybrid views are proposed, and below is my version called the dual-aspect model of properties.

The basic idea goes like this. Each property can be represented by a two-dimensional coordinate, in which one axis represents the quiddity of the property, and the other axis represents the causal powers of the property. The generation of any such coordinate is both brute and contingent (in other words, properties are not necessarily disposed). Furthermore, unlike dispositional essentialists or categorical essentialists, I tentatively suggest that a property is identified and individuated by the dual-aspect coordinate (i.e., its quiddity and its causal profile taken together), rather than the quiddity alone nor the causal profile alone (more reasons later). Three more clarifications to distinguish my proposal from other versions of hybrid views.

First, my proposal is different from the Dual-Aspect Account of properties developed by Joaquim Giannotti (2021), despite our theories' similar names. Giannotti argues that *dispositionality* and *qualitativity* are two aspects of *beings of properties*, and the notion of aspect is primitive such that

> 1) aspects are not any kind of entities, 2) all aspects of a property are both intraworldly and interworldly determined by the nature of the property itself (where 'world' means possible worlds), 3) aspects can be considered through an epistemic act of partial consideration (i.e., abstraction), and 4) most crucially, aspects of a property supervene upon the property (Giannotti 2021, pp.

615–616)[15].

It follows from Giannotti's account that properties are ontologically structureless entities and aspects are no addition to beings of properties. But I argue that my model implies properties are essentially structured and represented by two-dimensional coordinates. Second, my proposal is different from powerful qualities view (aka identity theory of powers). Unlike friends of powerful qualities view, I disagree that *being qualitative* and *being dispositional* are two ways of *being a property* that have no real distinction, i.e., qualitative aspect and dispositional aspect are identical (Coates 2021, p. 8347). I must reject such identity so that the emergent property is irreducible to its emergent base (more details at the end of section 3). Third, my proposal partly resembles the compound view that all properties are mereologically composite entities consisting of dispositional parts and qualitative parts (tentatively proposed in Taylor 2018, pp. 1438–1439). In addition, I tentatively suggest that the qualitative aspect is the salient aspect of a property, or analogously, a quiddity anchors a property' identity. Meanwhile, the dispositional aspect only assistants the identification and individuation of a property. Now, instead of diving into objections and replies to my proposal about nature of properties, as my argumentative strategy permits, I would like to transfer to the demonstration of how this proposal helps Wilsonian emergentists[16].

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[15] The notion of supervenience is, very roughly, a notion with a modal force that characterizes the covarying, dependent and non-reducible relation between a set of higher level (i.e., supervenient) entities and a set of lower level (i.e., subvenient) entities.

[16] For a detailed discussion of objections to the compound view, my proposal's relative, see Livanios 2020, pp. 282–290. And I shall paraphrase Livanios' conclusion that the "upshot of the preceding

Granted the dual-aspect model of properties, it is natural to follow up by asking what the relation between an emergent property's quiddity and its emergent base's quiddity is. Consider two properties (S and P) in the diagram below:

Diagram 1. An emergent property, S, and its emergent base, P

With respect to the relation between S's quiddity and P's quiddity, I tentatively suggest that Wilsonian emergentists may answer by two conditionals. If S is weakly emergent from P, then S's quiddity is a duplicate of P's quiddity. And if S is strongly emergent from P, then S's quiddity is completely independent of—and in no way generated/derived from—P's quiddity. I think, for Wilsonian emergentists, there are two obvious merits coming from a line of response as above.

The first merit is to offer a qualitative distinction between weak and strong emergence. Besides the quantitive distinction (Wilson 2021, p. 74), the dual-aspect model of properties also suggests that whether a property is weakly (or strongly) emergent should depend on whether its shares a (purely) qualitative quiddity with its emergent base. This distinction seems attractive in two aspects. For one thing, given Wilson's

discussion is that none […] offers a knockdown argument against [the compound view]" (2020, p. 289).

claim that "the base feature is physicalistically acceptable" (2021, p. 75), since a mental property's quiddity is identical with the quiddity of its emergent base (i.e., a physical quiddity) and is therefore physicalistically acceptable, the quiddity-based distinction can further justify why Wilsonian weak emergence is compatible with physicalism. For another thing, since a strongly emergent property's quiddity is entirely novel, is totally independent of, and is therefore not grounded in its emergent base's quiddity, the quiddity-based distinction also characterizes the robust novelty of strongly emergent entities that some emergentists demand[17]. And since the strongly emergent property's quiddity (i.e., a mental quiddity) is not grounded in its emergent base's quiddity (i.e., a physical quiddity), this distinction also explains why strong emergence is incompatible with physicalism (as suggested in Wilson 2021, pp. 46–53) by virtue of violating the No Fundamental Mentality constrain. I handle one potential complaint before proceeding. Some people may dislike the idea that a weakly emergent mental property shares a physical quiddity with its physical base. I have two disclaimers. First, having a physical quiddity does not imply that the mental property is reducible to its physical base (as I will argue in the subsequent paragraph). Second, the alleged counterintuitiveness does not come from my model but from the physicalist position itself, according to which every existence is ultimately physical. Quiddities are of course no exception.

The second merit is to defend the non-reductiveness of emergent-level properties.

[17] Most famously required by some British emergentists, see Alexander 1920, vol. 2, pp. 45–47, and Morgan 1923, pp. 9–10 for details

I have shown how dispositional essentialism fails to satisfy the ontological non-reductiveness desideratum in section 2. Similarly, given that a weakly emergent property inherits its quiddity from its emergent base, categorical essentialism also fails to satisfy the ontological non-reductiveness desideratum. Likewise, given either of two lines of reasoning above, and provided that a property's quiddity is identical to its causal profile, powerful qualities view again fails to satisfy the ontological non-reductiveness desideratum. I think, by far, the most promising defense of the non-reductiveness of emergent-level properties shall appeal to my proposal, that a property is identified and individuated by the dual-aspect coordinate (i.e., its quiddity and its causal profile taken together). For illustration, still consider two properties, S and P, from diagram 1. S weakly emerges from P, which implies that their quiddities (represented by vertical axes of two coordinates) are identical and their causal profiles (represented by horizonal axes of two coordinates) stand in an inclusive relation. However, horizonal and vertical axes taken together, there are two distinct coordinates: S is therefore irreducible to P.

4 *Ceteris Paribus* Emergent-level Laws

In section 4, I will illustrate how the dual-aspect model of properties explains the difference in logical form between emergent-level laws and base-level laws. Before that, I make four remarks.

First, the notion of law in my conception is, on top of everything, a scientific notion. Scientific laws are expressions that scientists believe to be general rules of their

subjects of research. This scientific conception of laws are not necessarily laws (of nature) that metaphysical theories suggest. For distinction, I call laws accepted by scientists while rejected by (some) metaphysicians "law-like expressions".

Second, from a physicalistic view, I do not distinguish base-level laws and laws of physics. But are laws of physics also laws of nature? I think physicalism in Wilson's (and my) conception is insensitive to this question. The core component of Wilson's conception of physicalism is the No Fundamental Mentality constrain that "attaches to […] entities and features which are the targeted subject matter of physics" (Wilson 2021, p. 23). That being said, the Wilsonian conception of physicalism does not require laws of physics to be laws of nature.

Third, the notion of fundamentality should not confuse with the notion of fundamental laws. To say something is fundamental is to say that thing is ontologically primitive. While to say a law is a fundamental law is to say that law cannot be reduce to other laws. For instance, laws of inorganic chemistry are not fundamental laws because they can be reduced to laws of atomic physics: anything that can be explained by laws of inorganic chemistry can be explained by laws of atomic physics. In a word, the notion of fundamental laws does not entail the notion of fundamentality. Fundamental laws in principle could be ontologically derivative (as opposed to primitive).

Fourth, there are two requirements at minimum for the explanation I intend to give[18]. Firstly, it should (at least) be reasonable and psychologically compelling.

[18] As Psillos 2007, p. 442 summarizes, these requirements arguably apply to any particular theory of

Secondly, it should be viable and produce truth (i.e., the explanans entails the explanandum). Besides, I wish that my line of explanation can also accommodate Wilson's pursuit of theoretical neutrality to the nature of laws (of nature) (2021, p. 18).

In spite of the neutrality I advertise, I think Wilsonian emergentists still have to presuppose something about laws, or law-like expressions (which may or may not be laws of nature). That is, usually, expressions appearing in the form that "universally/*ceteris paribus*, all Fs are Gs" (arguably the orthodox form of laws) can also be stated as "universally/*ceteris paribus*, if something has disposition $D_{(F, G)}$ and instantiates property F, then it instantiates property G". In other words, many laws (or law-like expressions) can be stated in terms of dispositions. This idea is sometimes known as the Nagel-style conception of laws[19]. I will justify this claim in section 4.1.

4.1 Scientific Laws, Dispositions and Laws of Nature

The claim, that many laws (or law-like expressions) can be stated in terms of dispositions, is countenanced by facts about formulations of laws in sciences like physics, chemistry, and biology. Only to mention three examples:

> *Newton's First Law*. An object at rest stays at rest and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force. (This law can be stated in terms of an object's

explanation.

[19] The Nagel-style conception of laws is basically the view that laws are "expressions of the powers and susceptibilities of concrete entities" (cf. Lewtas 2013, p. 539).

disposition to move or to stay at rest.)

Pauli's Exclusion Principle. One orbital can be occupied by at most two electrons whose spins must be anti-parallel. (This law can be stated in terms of an electron's disposition to spin or jump.)

Mendel's Law of Independent Assortment. The alleles of two different genes get sorted into gametes independently of one another. (This law can be stated in terms of an allele's disposition to separate.)

Examples above hopefully show how the claim could be true. In what follows, I will show why this claim is metaphysically innocent with respect to different theories of the nature of laws (of nature). And I divide conceptions of the nature of laws (of nature) into two groups: Humeanism and non-Humeanism.

Humeanism about the nature of laws (of nature) is, basically, the view that laws of nature are just patterns (or ways of describing patterns) in the mosaic of events—i.e., the intrinsic physical state of each spatial-temporal point (or each point-like object) and spatial-temporal relations between those points (Bhogal 2020, p. 1). As the famous quotation from David Lewis goes:

> "Humean supervenience […] is the doctrine that *all there is to the world is a vast mosaic of local matters of particular fact*, just one little thing and then another. ... We have geometry: a system of external relations of spatiotemporal distance between points. ... And at those points we have *local qualities*, *perfectly natural intrinsic properties*, which need nothing bigger than a point at which to be instantiated. For short: we have *an arrangement of qualities*. And that is all. There is no difference without difference in the arrangement

of qualities. *All else supervenes on that*." (Lewis 1986, pp. ix–x, italics mine)

In short, Humean laws of nature are local, intrinsic, non-modal (i.e., non-dispositional) and non-nomic patterns (i.e., arrangements of qualities) of the mosaic.

It is not hard to see the disagreement: Humeanists allow for no primitive disposition in laws of nature, while dispositions are constitutive to the claim about laws I proposed. But the severity of the disagreement is arguably illusory. For illustration, consider the following diagram:

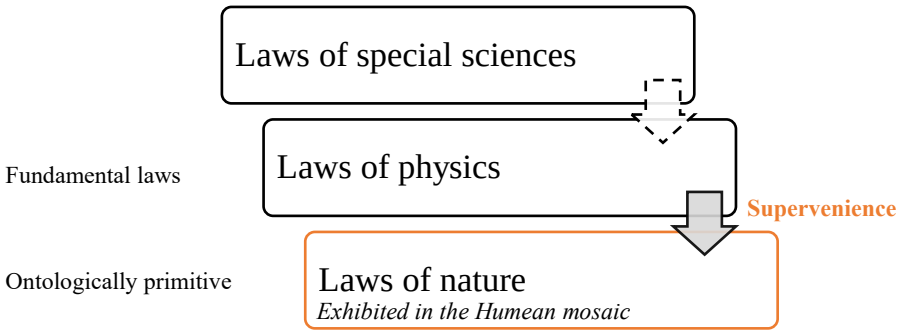

Diagram 2. The hierarchy of laws for Humeanists about the nature of laws of nature

Humeanists will assumably be happy to embrace this picture. Their commitment to non-dispositional laws of nature is accommodated. Physicalists in general have no reason to reject this picture, though some ambitious physicalist may dislike the idea that laws of physics are not laws of nature. But as I argued earlier, the Wilsonian conception of physicalism does not require laws of physics to be laws of nature. At the same time, emergentists will assumably also like the flexibility of this picture. On a strong emergentist view, laws of special sciences can be fundamental laws in the sense that they cannot be reduced to laws of physics. On a weak emergentist view, laws of special

sciences can supervene on laws of physics and are therefore not fundamental laws. To summarize, the claim that many laws (or law-like expressions) can be stated in terms of dispositions is compatible with Humeanism about the nature of laws (of nature). I consider one last concern before I transfer to my discussion of non-Humeanism. Some Humeanists (like David Lewis) also argue that fundamentally physical properties are essentially solely categorical. If so, dispositions of special scientific properties, if emergent, can only be strongly emergent. But as I have argued in section 3, strong emergence is incompatible with physicalism. Then there seems to be incoherence. I reply by arguing that the dual-aspect model should apply to all emergence-relevant properties, which implies fundamentally physical properties are *not* essentially *solely* categorical. That being said, a Wilsonian emergentist accepting my model can be a Humeanist about laws of nature, but *cannot* be a Humeanist about fundamental properties. Those two Humeanist positions, I think, are *prima facie* distinct and separable, and can be endorsed independently, unless further arguments are made (which do not exist now).

I now consider two predominant theories of non-Humeanism about laws of nature: the contingent nomic necessitation view[20] and the dispositionalist conception about laws. The contingent nomic necessitation view says something like this. Laws are fundamental second-order necessitating relations linking ordered pairs of first-order properties (i.e., sparse universals like *mass*, *charge*, and *spinning*) that are countenanced by

[20] Aka the Dretske–Tooley–Armstrong view of laws (Dretske 1977; Tooley 1977; Armstrong 1983). I particularly focus on Armstrong's version here.

scientific theories (Armstrong 1983, pp. 71–102). Formally put, let F and G two properties, a law can be written via an 'N' predicate as N <F, G>. Question: is the contingent nomic necessitation view compatible with the claim that many laws (or law-like expressions) can be stated in terms of dispositions? Probably yes. The key to an answer is how relation N should be understood. Armstrong's mysterious (and much-complained) relation N is, according to some interpretations (e.g., Heil 2016, p. 128), probably causal relation. If so, the contingent nomic necessitation view will imply that being an F *causally necessitates* being a G—a strengthened version of the claim I proposed[21]. Last but not least, consider the dispositionalist conception about laws, which says that laws flow from the dispositional essence of fundamental properties (i.e., potencies) that are simultaneously natural, metaphysical and scientific, and laws must hold in every possible world where relevant potencies exist (Bird 2007, pp. 43–65)[22]. This position initially entails a strengthened version of the claim I proposed. In fact, Bird discussed the relation between having a disposition $D_{(S,M)}$ and the law "All Ss are Ms" (2007, p. 152) which resembles the formulation "for any *x*, if F*x* and D$x_{(F, G)}$, then G*x*" that I suggest. But it is worth clarifying my disagreement with Bird: I reject that dispositionality is the sole essence of properties, and I am neutral to question whether a law must

[21] And of course, even if a Heil-style interpretation is false, I can still compromise by arguing that disposition-based scientific laws are merely law-like expressions supervening on laws of nature. Given my distinction at the beginning of section 4, I think a compromise as above is arguably not too hard a bullet to bite.

[22] Following Bird himself in *Introduction* of his 2007 book, I employ the term 'flow from' to describe the relation between laws and potencies, for comprehensibility. Nevertheless, in Bird's 2007 book, laws are logically deducible from potencies. See Bird 2007, p. 46 for details.

hold in every possible world where its constitutive disposition exists.

4.2 A Dual-aspect Model-Based Explanation

Now back to the question that how the dual-aspect model of properties explains why many emergent-level laws are *ceteris paribus* laws. For demonstration, consider the biological/special-scientific/emergent-level law: "*ceteris paribus*, the lack of vitamin C causes scurvy." Provided what I say in section 4.1, we may assume that this law can be stated in terms of (some) disposition(s). To see how, let's take a closer look at the story about being lack of vitamin C and having scurvy.

There are (at least) two narratives of the story: one biological, the other physical. The biological narrative goes like this. Cells (or functional parts of cells like ribosome or endoplasmic reticulum) of human body need vitamin C (aka ascorbic acid) to synthesize several kinds of proteins that are crucial to the construction of tissues, transplantation of nutrition, etc. The lack of vitamin C makes human body lack of those crucial proteins, along with which come symptoms of scurvy. By contrast, the (atomically) physical narrative goes like this. Various kinds of atoms (like carbon atoms, hydrogen atoms and oxygen atoms) stand in some fundamental physical relations (like spatial-temporal relation or force), on some conditions, the configuration of atoms exhibits some relatively stable structures, from which some higher-order properties (like *being a protein molecule* or *being a cluster of molecules that is biologically a cell*) emerge. Those atoms (or configurations of atoms) constantly interact with each other,

the physical properties they instantiate (like *being a carbon atom*, or *having such and such a rest mass*) are relatively stable, while the higher-order property emergent from the physical property may persist or cease to exist.

Bearing those two lines of narratives in mind, I now introduce some notations to be employed. Let A be the token biological property instantiated by a person's cells when that person is lack of vitamin C, and E be the token physical property from which A is weakly emergent. Let property B be the token biological property instantiated by a person's cells when that person has scurvy, and F be the token physical property from which B is weakly emergent. Let disposition $D_{(A, B)}$ be the token emergent-level disposition relevant to the emergent-level law at issue. Let *x* be an individual variable to which the law at issue applies. Consider this diagram for illustration:

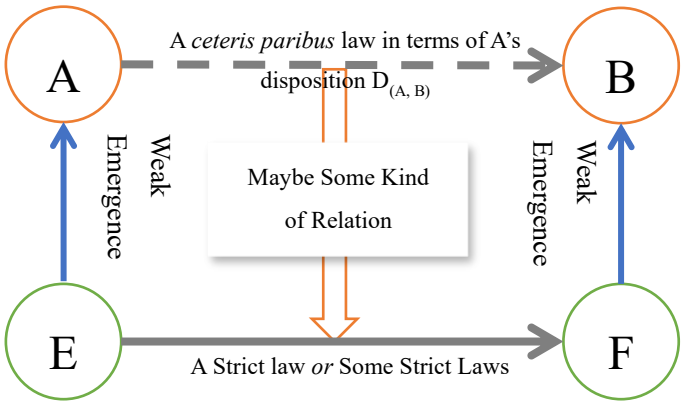

Diagram 3. Two biological properties, their physical emergent bases and laws in two levels

Given the claim that many laws (or law-like expressions) can be stated in terms of dispositions, the emergent-level law at issue can be stated as "*ceteris paribus*, for any *x*, if *x* is A and has disposition $D_{(A, B)}$, then *x* is B"[23].

[23] Correspondingly, at the base level, the process from E to F—two token physical properties of a

Considering the syntactical structure (i.e., a conditional) of this *ceteris paribus* emergent-level law, and given that one conjunct (i.e., *being A*) of the antecedent (a conjunction) is true, to explain why there are exceptions of *being A and not B* is to explain why *having disposition* $D_{(A, B)}$ can be false when *being A* is true. Below is a dual-aspect model-based explanation. As is seen, *being A* can be considered as token property A's quiddity, and $D_{(A, B)}$ is one possible causal power of A. The explanandum that *having disposition* $D_{(A, B)}$ can be false when *being A* is true can be explained by two (interconnected) explanantia. The first explanans is the dual-aspect model's basic posit that the combination of any quiddity and causal powers are contingent. Yet I do not want a stipulative explanation. I think it is reasonable to ask why the posit is so-made. And I thereby provide the second (also ultimate) explanans: the bruteness of emergence. As I said in section 2, emergence is brute: how and why sometimes the token emergent property inherits a particular causal power is brute, and how and why sometimes the token emergent property does *not* inherit a particular causal power is also brute. That being said, the brute fact that some tokens of property A do not have the causal power $D_{(A, B)}$ after emergence explains why there are instances of *being A and not B*. To generalize this line of reasoning, the point that a property's quiddity and its causal powers are brutely and contingently combined explains why many emergent-level laws are *ceteris paribus* laws.

complicated particles configuration—should also exhibit (or be governed by) at least one law of physics (as taken for granted) that runs as "universally, for any *x*, if *x* is E and has disposition $D_{(E, F)}$, then *x* is B".

One residual concern. Still consider the case of "the lack of vitamin C causes scurvy". For instances in which a person is lack of vitamin C but does not have scurvy, the base-level physical property is instantiated by a configuration of particles that enables particular individuals' cells to synthesize crucial proteins without vitamin C. While for instances in which the lack of vitamin C causes scurvy, the base-level physical property is instantiated by a configuration of particles that forbids individual persons' cells to synthesize crucial proteins without vitamin C. Those two configurations seem to be (quite) different. Question: why are those two physical properties not two different types? I think there are two reasons. First, particle configurations instantiating those two physical properties exhibit one similar type of structure, i.e., the structure of particles in cells from similar human tissues. Arguably, the difference between those two configurations is, though biologically great, physically subtle. The difference is probably just different numbers of configurating particles, longer/shorter spatial distances, etc. It is dubious whether differences as such are great enough to make different types. Second, differences between those two physical properties are intuitively (much) smaller than differences between the physical base of pain in human brains and the physical base of pain in octopus foots or in Martian organisms. Since many physicalists believe those seemingly diverse physical bases can be considered as different tokens that multiply realize the mental property pain (e.g., Lewis 1980, p. 216; Heil 1999, p. 194), I think there are also good reasons to believe that those two physical properties at issue are just different tokens of one same type.

5 Conclusion or Something Close Enough

A quick review of my work thus far. I first argue that the alleged neutrality of Wilson's 2021 account of weak emergence to the nature of properties is overstated (in section 2). Later, to modify Wilson's account, I propose the dual-aspect model of properties which helps Wilsonian emergentists i) qualitatively differentiate weak emergence from strong emergence, ii) defend the non-reductiveness of emergent-level properties (in section 3), and iii) explain why many emergent-level laws are *ceteris paribus* laws (in section 4). Till this point, my conclusion is presupposing the dual-aspect model of properties improves a Wilsonian conception of (weak) metaphysical emergence.

In closing, I wish to clarify several points. My focus in this project, to start, is Wilson's account of weak emergence. I shall consequentially remain silent to the evaluation of strong emergence. Besides, whereas I directly target on the formulation of non-reductive physicalism, my conclusion in principle generalizes to composed objects, freedom in a deterministic world and collective agency, if they are also (weakly) emergent entities as the last three chapters of Wilson 2021 explore. In the end, considering the notion of metaphysical emergence is fashionably popularized (at least I hope so), at this moment, with respect to the prospect of the industry, it is too hasty to draw any conclusion. Nevertheless, there is one thing that is predictable: Wilson's account of (weak) emergence will be discussed for a considerable length of time in lights of its originality, ambition, and alas, controversy.

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